KIEU STUDIO · GITHUB RADAR · EPISODE 36 · CHEAT SHEET

auto-ontology

A context layer so agents answer data questions with the right numbers: terms, relationships, checkable SQL.

github.com/NVIDIA/auto-ontology

+9,327	10,534	Apache-2.0
stars in 7 days · ≈ 1332 stars/day · Trending lists no 30-day figure	total stars · GitHub API	license · created Apr 27, 2026

NVIDIA Auto Ontology (Apache-2.0) builds a context layer from enterprise metadata: terms, SQL attributes and table relationships. People certify it, then an agent answers questions with the SQL and the steps it took; it is served to agents through MCP, a command line and an API.

How it works in six steps

1. **Read the data sources** — schemas, sample values
2. **Name the terms** — map them to columns
3. **Infer relationships** — keys and bridge tables
4. **People certify it** — test, fix, approve
5. **Agent finds the terms** — embedding search
6. **Write SQL, verify** — answer with SQL and steps

Quick start

```
# Clone the repo
git clone https://github.com/NVIDIA/auto-ontology.git
cd auto-ontology

# Create the env file
cp .env.example .env
# add DEFAULT_MODELS_API_KEY, change every password

# Run the whole stack
docker compose up -d --build

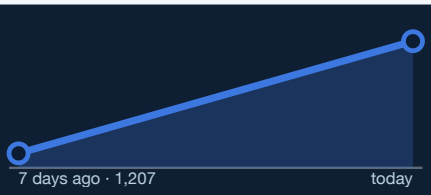
# Open the web UI
http://localhost:3000 → Settings → Semantic Compilation

# Needs: Docker (Compose v2), uv and an NVIDIA API key (nvapi-...) from build.nvidia.com
```

Key commands / features

- **`docker compose up -d --build`** — build and run the services
- **`uvx ... auto-ontology-mcp`** — the MCP server (read-only)
- **`check_answerable`** — MCP tool: does the ontology cover it
- **`ask_question`** — MCP tool: answer, SQL, rows
- **`search_terms`** — MCP tool: find terms

Growth



Two real points: the start of the window and today, axis from zero. The gain is as reported by GitHub Trending; we did not rebuild the star history.

Why this repo exists

- Agents answer fast, but a number can be wrong if they do not know where data lives, what it means and where it came from.
- A shared context layer gives every agent the same definitions and makes every answer checkable.

What to use it for

- Ask enterprise data questions in plain language
- Give agents consistent definitions through MCP
- Check answers through the SQL and steps shown
- Certify terms and SQL attributes

Think twice before you use it

- Uses NVIDIA NIM endpoints (hosted or self-hosted) by default and needs an API key.
- Compose publishes Postgres, pgAdmin and the web app: replace every default password.
- Keep ports 3001 and 3002 on a private network: the ingestion service does not authenticate.
- The project is not accepting outside contributions; certify definitions before you trust a number.

Sources: the repo's README, https://github.com/NVIDIA/auto-ontology. Stars: GitHub API; gains: GitHub Trending (as reported by Trending), snapshot Nov 9, 2026; stars measure attention, not quality: re-check before relying on them. Summarized in Kieu Studio's own words; commands copied from the README; no third-party logos or screenshots. **Disclosure:** drafted with AI assistance; the voice and presenter in the videos are AI-generated. Not investment advice, and not endorsed by Kieu Studio. **kieu.app**